

	DEPARTMENT OF IT		
	FACULTY OF Computing		
MODULE OUTLINE			
Module Name	Programming Applications and Frameworks		
Module Code	IT3030	Version No.	2024-1
Year/Level	3	Semester	1
Credit Points	4 Credits		
Pre-requisites	N/A		
Co-requisites	N/A		
Methods of Delivery	Lectures (face to face)	2 Hours/Week	
	Tutorials	1 Hour/Week	
	Laboratories	4 Hours/Fortnight	
Course web site	http://courseweb.sliit.lk		
Date of Original Approval	January, 2017		
Date of Last Approval	January, 2024		
Date of Next Review	January, 2029		
MODULE DESCRIPTION			
Aim	This module discusses the contemporary concepts, technologies, frameworks/libraries, tools, and industry best practices used to engineer enterprise systems. Students will be exposed to both standard web-based applications development as well as the rich web-based applications development, using heavy client-side components.		
Learning Outcomes	At the end of the module, the student will be able to:		
	LO1:	Apply the basic concepts of frameworks	
	LO2:	Evaluate and apply available industry standard software development practices	
	LO3:	Create web-based applications using Java frameworks	
	LO4:	Apply the REST architectural style for web services	
	LO5:	Create client-side components of full stack web-based applications using JavaScript frameworks/libraries/plugins	
Assessment Criteria	Assessment Description:		
	During the semester there will a mid-term examination, a practical test, assignments, and a final exam.		

	Assessment Activities:		
	Mid-Semester examination	20%	L01, L02, L04
	Assignments/Laboratories	30%	L01-L05
	Practical Test	10%	L05
	End of Semester Examination	40%	L01, L02, L04
	Total	100 %	
Estimated Student Workload	Contact Hours		
	Lectures (face to face)	28 hours	
	Tutorials	14 Hours	
	Laboratory	14 hours	
	Time Allocated for Assessments		
	Continuous Assessments	02 Hours	
	Mid-Semester Examination	01 Hour	
	End of Semester Examination	02 Hours	
	Reading and Independent Study	139 Hours	
	Total	200 Hours	
Module Pass Requirements	To pass this module, a student needs to obtain a pass mark (45%) for each of “Continuous Assessments” and “End of the Semester Examination” components, which would result in an overall mark that would qualify for a “C” grade or above.		
Learning Resources	Recommended Texts Ian Gorton, "Foundations of Scalable Systems," O'Reilly Media, Inc., 2022.		
	Software Spring Boot Reference Documentation		
CONTENTS OF THE MODULE			
1.	Introduction to Frameworks 1 (5 hours)		L01
	• Programming paradigms • Software runtime architecture • Development tools • Frameworks vs libraries vs plugins		
2.	Industry Best Practices (5 hours)		L02

	• Version controlling with Git • CDN • Virtualization • Code quality • Dependency/package management • Build tools											
3.	Using Java basics for Web-based applications development (5 hours) Distributed systems • Distributed system types • Distributed systems architectures and styles • Communication in Distributed systems – RMI/CORBA/SOAP/REST, XML/JSON • Java Web Application Development JSP, Servlets	L03										
4	Using Java frameworks (10 hours) • SOAP web services – JAX-WS • Data persistence – ○ Files VS DB, DB types (un/semi/fully structured), parametric statements	L03										
5	RESTful web service development (10 hours) • REST architectural style • RESTful web services using server-side MVC	L04										
6	Introduction to client-side development with JavaScript frameworks (10 hours) • Core features of a typical JavaScript Framework • jQuery • Rich Web-based Applications, Delta-Communication, AJAX	L05										
7	Using AngularJS (11 hours) • Introduction to AngularJS Framework. • Angular and typescript with client-side MVC • Modules, controllers, Directives, Scope Object, Router	L05										
LEARNING OUTCOMES TO PROGRAM OUTCOMES MAPPING												
	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012
L01					X	X						
L02		X			X							
L03			X		X	X						
L04			X		X	X						
L05			X		X	X						
P01	Apply knowledge of mathematics, statistics, basic sciences, and computer science to solve complex problems related to the computing domain.											
P02	Identify, formulate, and analyze complex problems related to the computing domain											

	by using established approaches in computing while recognizing their limitations, reviewing technical literature, and interpreting the results of experiments designed using appropriate research methods.
PO3	Design systems, components, or processes to meet desired needs within realistic constraints.
PO4	Apply reasoning informed by contextual knowledge in the area of information technology to assess the impacts of IT solutions in a global, economic, environmental, societal, legal, cultural, and health & safety context.
PO5	Design, implement and deploy IT solutions using industry best practices including continuous improvements, quality assurance and mitigating security risks.
PO6	Create, select, and apply appropriate computing tools, techniques and resources necessary to solve IT related real world problems
PO7	Apply broad knowledge of sustainable development concepts and practices required for developing a solution in the information technology domain.
PO8	Apply ethical principles and commit to professional ethics, responsibilities, and norms in the field of Computing
PO9	Function effectively within a multidisciplinary and multicultural team which promotes equality, diversity, and inclusion.
PO10	Communicate on computing activities with a wide variety of audiences by means of appropriate designs, plans, reports, presentations, and instructions.
PO11	Apply broad knowledge of management principles, commercial context, change management, relevant legal matters including intellectual property rights and risk mitigation to manage projects in a multidisciplinary context.
PO12	Plan and record self-learning and continuous professional development as the foundation for lifelong learning.
GENERIC INFORMATION	
Any type of plagiarism is not allowed.	
Plagiarism: Academic honesty is crucial to a student's credibility and self-esteem, and ultimately reflects the values and morals of the Institute as whole. A student may work together with one or a group of students discussing assignment content, identifying relevant references, and debating issues relevant to the subject. Plagiarism occurs when the work of another person, or persons, is used and presented as one's own.	
END OF MODULE OUTLINE	